

Wilson loops in supersymmetric gauge theories: 1

Plan

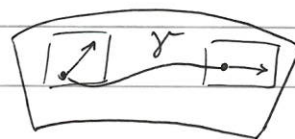
1. WL in gauge theories: - Defn
 - Interparticle potential
2. SUSY WL: - $N=4$ SYM theory in 4d
 - Maldacena-Wilson loop
 - Examples
 - Perturbation theory
3. Circular WL: - Exact vev from PT
 - Localize the circular WL
 - Exact vev from matrix model

Trancanelli, school on SUSY localisation, ICTP website
 Cremonesi, intro to localisation, POS Sissa

1. WL in gauge theories

1.1 Definitions

$$x^\mu = x^\mu(\tau), \quad \tau \in [\tau_0, \tau_1]$$



assume metric, connection in one $T_p M$, draw a curve s.t. they then belong to the same $T_p M$

- Parallel transport equation

$$\dot{x}^\nu \nabla_\nu V^\mu = 0 \equiv \dot{x}^\nu (\partial_\nu V^\mu + \Gamma^\mu_{\nu\lambda} V^\lambda)$$

$$(*) \quad \frac{d}{d\tau} V^\mu(\tau) = -(\Gamma^\mu_{\nu\lambda} \dot{x}^\nu V^\lambda)(\tau), \quad V^\mu(\tau_0) = \bar{V}^\mu$$

$$V^\mu(\tau) \equiv U^\mu_{\nu}(\tau, \tau_0) \bar{V}^\nu$$

parallel propagator

$$\text{eqn for } U: \begin{cases} \frac{d}{d\tau} U^\mu_{\nu}(\tau, \tau_0) = -\Gamma^\mu_{\sigma\lambda}(\tau) \dot{x}^\sigma(\tau) U^\lambda_{\nu}(\tau, \tau_0) \\ U^\mu_{\nu}(\tau_0, \tau_0) = \delta^\mu_{\nu} \end{cases}$$

$$U^\mu_{\nu}(\tau, \tau_0) = P \exp \left(- \int_{\tau_0}^{\tau} \Gamma^\mu_{\sigma\lambda} \dot{x}^\sigma d\tau \right)$$

P: path ordering operator

Properties : 1) $U \in G$: $\psi(x) \rightarrow \psi(x) = U(x) \psi(x)$ (quark)
 (connection) $A_\mu(x) \rightarrow A_\mu(x) = U(x) A_\mu(x) U^\dagger(x) - i \partial_\mu U(x) U^\dagger(x)$

$$U_P(y, x) \rightarrow U_P(y, x) = U(y) U_P(y, x) U^\dagger(x)$$

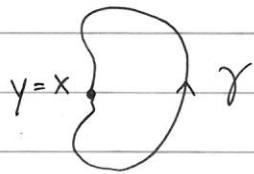
$$U_P(y, x) \psi(x) \rightarrow \underline{U(y)} U_P(y, x) \underline{\psi(x)}$$

related to quark in other point

$$2) U_P(y, x) = U_P(y, z) U_P(z, x)$$

we want gauge invariant operators !

- Identify a closed curve called loop



$$U_P(x, x) \rightarrow U(x) U_P(x, x) U^\dagger(x)$$

$$\text{tr}(U_P(x, x)) \rightarrow \text{tr}(U_P(x, x) U^\dagger(x) U(x)) = \text{tr}(U_P(x, x))$$

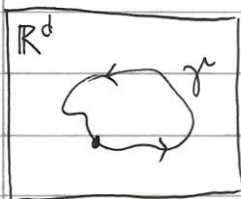
Wilson loop: $W_R[\gamma] = \text{tr}_R U_P(x, x)$

$$= \text{tr}_R \text{Pexp} \left(i \int_\gamma A_\mu dx^\mu \right)$$

- non-local, gauge invariant, depends on γ
- WL is γ independent iff $F_{\mu\nu} = 0$
- depends on R (repn) we could parallel transport something else : tr_R means you need to consider different combination of eigenvalues [Armoni, Shifman, Veneziano paper looking at different $\rightarrow 0307097$]
 representation ($R = \square \sim \square$ for $SU(3)$; $\tilde{q}^{ij} = \epsilon^{ijk} q^k$)

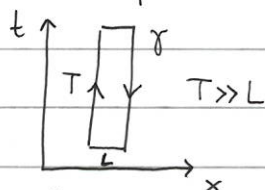
Wilson Loops in supersymmetric gauge theories: 2

Wilson loop operator



$$W_R(\gamma) = \text{tr}_R U_P(x, x) = \text{tr}_R P \exp \left(i \int_{\gamma} A_{\mu} dx^{\mu} \right)$$

1.2 Interparticle potential



$$S_{\text{maxwell}} = \int d^4x \left(\frac{1}{4e^2} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2e^2} (\partial^{\mu} A_{\mu})^2 \right)$$

$$F_{\mu\nu} = \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}$$

euclidean space

$$\langle A_{\mu}(x) A_{\nu}(y) \rangle = \Delta_{xy} \delta_{\mu\nu} = \frac{e^2}{4\pi^2 (x-y)^2} \delta_{\mu\nu}$$

$$W[\gamma] = e^{i \int_{\gamma} A_{\mu} dx^{\mu}}$$

• 2 static p.cle and antip.cle

γ = rectangle that includes $x^{\mu}(T_1) = (T_1, 0, 0, 0)$

$$x^{\mu}(T_2) = (-T_2, L, 0, 0)$$

$$\langle W[\text{rect}] \rangle = \frac{1}{Z} \int \mathcal{D}A e^{-S_{\text{maxwell}} + i \int_{\gamma} A_{\mu} dx^{\mu}} = \frac{1}{Z} \int \mathcal{D}A e^{-S_{\text{maxwell}} + i \int_{\gamma} A_{\mu} dx^{\mu} + \int d^4x \left(-\frac{1}{4e^2} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2e^2} \partial^{\mu} A_{\mu} \partial^{\nu} A_{\nu} - A_{\mu} J^{\mu} \right)}$$

Peskin ch 9

$$= \frac{1}{Z} \int \mathcal{D}A e^{-S_{\text{maxwell}} + i \int_{\gamma} A_{\mu} dx^{\mu} + \int d^4x \left(-\frac{1}{4e^2} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2e^2} \partial^{\mu} A_{\mu} \partial^{\nu} A_{\nu} - A_{\mu} J^{\mu} \right)}$$

$$= \frac{1}{Z} \int \mathcal{D}A e^{-S_{\text{maxwell}} + i \int_{\gamma} A_{\mu} dx^{\mu} + \int d^4x \left(\frac{1}{2e^2} A^{\mu} \square A_{\mu} - A_{\mu} J^{\mu} \right)} = \frac{Z[J]}{Z[0]} \sim e^{-V(L)T}$$

difference of ground state ~~of~~ of full vacuum or w/ 2 quarks.

Gaussian in A

$$\begin{aligned} (*) &= \exp \left(\int d^4x d^4y -\frac{1}{2} J^{\mu}(x) \Delta_{xy} J^{\mu}(y) \right) = \text{EXERCISE} = \\ &= \exp \left(\int_0^T d\tau_1 \int_0^{-T} d\tau_2 \frac{-e^2}{4\pi^2 [(T_1 + T_2)^2 + L^2]} \right) = \exp \left(\frac{e^2}{4\pi} \frac{T}{L} \right) \end{aligned}$$

$$\rightarrow V(L) = \frac{-e^2}{4\pi L}$$

Perimeter law: if $\langle W[\text{rect}] \rangle \sim e^{-kP}$ ^{perimeter} $\rightarrow V(L) = \text{const.}$ Coulomb phase

Area law: if $\langle W[\text{rect}] \rangle \sim e^{-\sigma A} \rightarrow V(L) \propto L$ Confining phase

2.2 Maldacena-Wilson loop

[Maldacena 98]

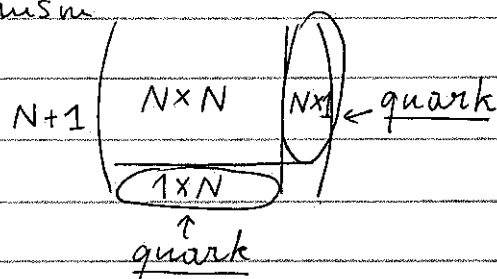
susy generalization of bosonic WL

• we don't have a quark in $N=4$ SYM, need a trick:

introduce quarks via the Higgs mechanism

$$U(N+1) \rightarrow U(N) \times U(1)$$

quark w/ large mass, usually called W boson

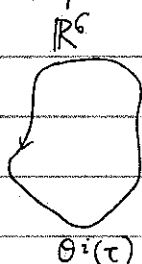


• ansatz for WL, include ϕ_i :

- A and ϕ belong to the same multiplet

- dim reduction $W[\gamma_{(10)}] = \text{tr}_R \text{Pexp} \left(i \int_{\gamma_{(10)}} A_M dx^M \right)$

$$A_M dx^M \rightarrow A_\mu dx^\mu + \phi_i dx^i \quad X^M(\tau) \rightarrow (x^\mu(\tau), \theta^i(\tau))$$



- also bc variation of $\delta_\epsilon \phi$ may cancel $\delta_\epsilon A$

$$W_R[\gamma, \theta] \equiv \frac{1}{d(R)} \text{tr}_R \text{Pexp} \left[i \int_\gamma dt A_\mu \dot{x}^\mu - i |\dot{x}| \theta^i \dot{\phi}_i \right]$$

generalized connection

• depends on $\gamma, \theta^i(\tau), R$

• $|\dot{x}|$ ~~reparam~~ reparametrisation invariance $\tau \rightarrow f(\tau) = \tau', A = A_\mu \dot{x}^\mu(\tau)$

$$\langle W \rangle = \frac{1}{d(R)} \text{tr} \mathbb{I} + \mathcal{O}(g^2) = 1 + \mathcal{O}(g^2)$$

normal.

• because we are in Euclidean scalar part x^i (?)

Supersymmetric: invariant under combination of all f. ops $Q \& S$

WL are locally $\frac{1}{2}$ BPS for any γ, R and $\theta^i(\tau) \theta_i(\tau) = 1$

$$0 = \delta_\epsilon W \Rightarrow \delta_\epsilon A_\mu = 0 = \epsilon (i \Gamma_\mu \dot{x}^\mu + |\dot{x}| \Gamma_i \theta^i) \Psi \stackrel{!}{=} 0 \quad (\text{swap } \epsilon \leftrightarrow \Psi)$$

$$(i \Gamma_\mu \dot{x}^\mu + |\dot{x}| \Gamma_i \theta^i) \epsilon(\tau) = 0 \quad \text{Local BPS equation}$$

local because $\dot{x}, |\dot{x}|$ and θ depend on (τ)

BPS: supersymmetric

ensure to ensure solutions exist $\rightarrow \theta^2 = 1$

Have to make this global

$$\rightarrow P_\pm = \frac{1}{\sqrt{2}} \left(1 \pm i \frac{|\dot{x}|}{|\dot{x}|} \Gamma_\mu \Gamma_i \theta^i \right)$$

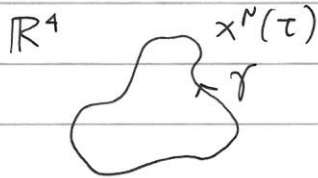
projectors

$$\text{BPS eq} \rightarrow P^- \epsilon = 0 \rightarrow \epsilon = P^+ \bar{\epsilon}$$

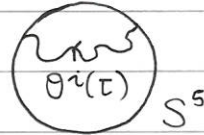
16 supercharges
32 supercharges

Wilson Loops in supersymmetric gauge theories: 3

$$W_R[\gamma, \theta] \equiv \frac{1}{d(R)} \text{tr}_R \text{Pexp} \left[i \int_{\gamma} \underbrace{(A_{\mu} \dot{x}^{\mu} - i |\dot{x}| \theta^i \phi_i)}_{A = A_{\mu} \dot{x}^{\mu}} d\tau \right]$$



$\mathbb{R}^6$



$$\theta^i(\tau) \theta^i(\tau) = 1$$

$\rightarrow \frac{1}{2}$ BPS (locally) $\forall \tau \quad [i \Gamma_{\mu} \dot{x}^{\mu}(\tau) + |\dot{x}(\tau)| \Gamma_i \theta^i(\tau)] \epsilon(\tau) = 0$
 Can WL preserve SUSY rigidly / globally i.e. $\forall \tau \quad \epsilon(\tau) = \epsilon$?

2.3 Example

Complete Classification [Dymarsky Pestun 2009]

- Straight line, $\frac{1}{2}$ -BPS

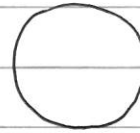
$$x^{\mu} = (t, 0, 0, 0) \quad \theta^i(\tau) = (0, 0, 0, 1, 0, 0)$$

EXERCISE: bosonic + fermionic symm

- Circular (DGRT) loop, $\frac{1}{2}$ BPS

$$x^{\mu}(\tau) = (\cos \tau, \sin \tau, 0, 0)$$

$$\theta^i(\tau) = (0, 0, 1, 0, 0, 0)$$



EXERCISE

[Bianchi, Green, Kovacs '02, Section 2]

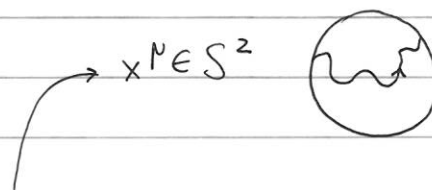
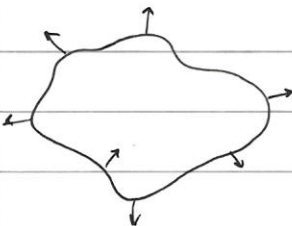
[Drukker, Gaiotto, Ricci, Trnconelli '07]

we can^{also} vary θ

- Zarembo loops in $\mathbb{R}^4$ [2002]

$$\theta^i = M^i_{\mu} \frac{\dot{x}^{\mu}}{|\dot{x}|}$$

$$M^i_{\mu} M^i_{\nu} = \delta_{\mu\nu}$$



localization
[Pestun '09]

- DGRT loops in $S^3 \subset \mathbb{R}^4$

exp value $\neq 1$

SU(2) right invariant quarks

$$x^{\mu} x^{\mu} = 1$$

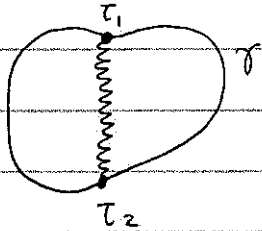
$$i = 1, 2, 3$$

$$|\dot{x}| \theta^i = \frac{1}{2} \sigma_i^R M^i_i, \quad |\dot{x}| \theta^{i+3} = 0$$

$$M^i_{\pm} M^j_{\pm} = \delta_{ij}$$

$$I = 1, \dots, 6$$

$$= 1 + \frac{g^2 N^2}{8\pi^2} \frac{C(R)}{d(R)} \int dT_1 dT_2 \underbrace{\left(\frac{-\dot{x}_1 \cdot \dot{x}_2 + |\dot{x}_1| |\dot{x}_2| \theta_1 \cdot \theta_2}{(x_1 - x_2)^2} \right)}_{\text{scalar couplings}} + O(g^4)$$



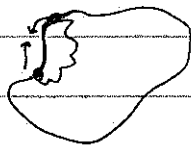
"1-loop propagator"

$$\text{wavy line} = \text{gluon} + \text{scalar}$$

in vevs there are both $\hat{P}$ and $\hat{T}$, do not override each other bc $\hat{P}$ acts on T^a and $\hat{T}$ acts on A^a_μ

• WL's vev is finite when γ is smooth, non self-intersecting and θ is smooth

1-loop



$$T_2 = T_1 + \epsilon, \quad \dot{x}(\tau) \cdot \dot{x}(\tau) = 1$$

$$\theta^i(\tau) \theta_i(\tau) = 1$$

$$\frac{1 + \epsilon + \epsilon^2 + \dots}{\epsilon^2} = 1 + O(\epsilon)$$

all lops conjecture [Drucker Gross 2000]

Circle loop $R = \square$

$$C(\square) = \frac{1}{2} \quad d(\square) = N$$

$$\langle W_\square[\text{circle}] \rangle = 1 + \frac{\lambda}{8} + O(\lambda^2) \quad g^2 N = \lambda$$

$$\text{1-loop prop} = \frac{1}{2} \frac{1}{8}$$

3. Circular WL

3.1 Exact vev from perturbation theory

$$\langle W_\square[\text{circle}] \rangle = 1 + \text{1-loop exchange} + \text{2-loop exchange} + \text{2-loop mercedes} + \text{1-loop bubble} + O(\lambda^3)$$

in t'Hooft lim $\left[\int \langle A A \rangle_{(0)} \int \langle A A A A \rangle_{(0)} \int \langle A A A A A A \rangle_{(0)} \int \langle A A \rangle_{(1)} \right]$
 $g^2, N^{\frac{1}{2}} \rightarrow 0 : \lambda = \text{const}$

$$= 1 + \frac{\lambda}{8} + \frac{\lambda^2}{192} + \underbrace{\text{divergence} + \text{divergence}}_{=0} + O(\lambda^3)$$

ladders

Wilson loops in supersymmetric gauge theories: 4

Recursion for A_n : $A_{n+1} = \sum_{k=0}^n A_{n-k} A_k$ $A_0 = 1$

[also  is allowed] Ladder = non

↑
also a
ladder

Solve the recursion $\rightarrow$ generating function

$$f(z) = \sum_{k=0}^{\infty} A_k z^k$$

Properties: $\begin{cases} f(z)^2 = \frac{f(z)-1}{z^2} \\ f(0) = 1 \end{cases}$

$$f(z) = \frac{1 - \sqrt{1 - 4z}}{2z}$$

$$= \sum_{n=0}^{\infty} \frac{(2n)!}{(n+1)! n!} z^n$$

$$\Rightarrow A_n = \frac{(2n)!}{(n+1)! n!}$$

$$\forall \lambda, N \gg 1 \quad \langle W_0(\text{circle}) \rangle = \sum_{n=0}^{\infty} \frac{1}{n!(n+1)!} \left(\frac{\lambda}{4} \right)^n$$

$$= \frac{2}{\sqrt{\lambda}} I_1(\sqrt{\lambda})$$

$$= \begin{cases} 1 + \lambda/8 + \lambda^2/192 + \dots & \lambda \ll 1 \\ \sqrt{\frac{2}{\pi}} \lambda^{-3/4} e^{\sqrt{\lambda}} (1 + \dots) & \lambda \gg 1 \end{cases}$$

exp constant

with holography

[Drukker, Gross 00] $\forall \lambda, \forall N : \frac{1}{N} L_{N-1}^1 \left(-\frac{\lambda}{4N} \right) e^{\lambda/8N}$

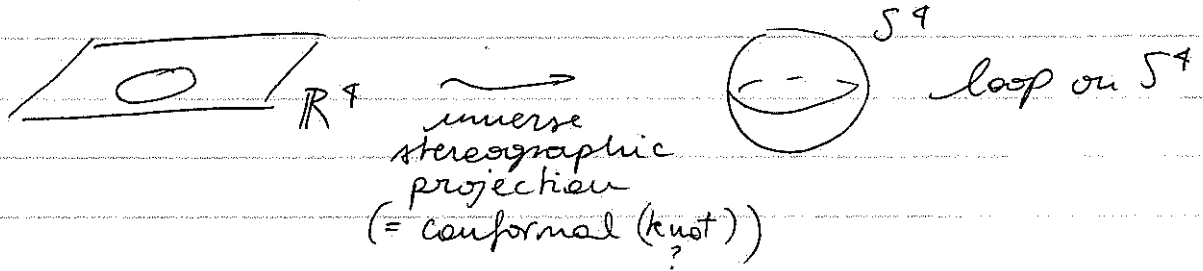
Laguerre

of course

This can also be used to compute expectation values of operators annihilated by δ .

Circle localization [Peshkin '07]

$N=4$ SYM in $\mathbb{R}^4$, circular loop



~~5484~~

Rem: δ_ϵ should be defined off shell $\rightarrow$ aux fields
 $K_i = 1, \dots, 7$

final expression: $\langle W_R[\text{circle}] \rangle_{\mathbb{R}^4} \overset{\text{conf. symm}}{=} \langle W_R[\text{circle}] \rangle_{S^4} =$

$$= \left\langle \frac{1}{d(R)} \text{tr}_R e^{2\pi M} \right\rangle_{\text{Gaussian matrix model}} =$$

$$= \frac{1}{Z} \int dM e^{-\frac{8\pi^2 N}{\lambda} \text{tr}_R M^2} \frac{1}{d(R)} \text{tr}_R e^{2\pi M} \cdot 1$$

Wilson Loops in supersymmetric gauge theories: 5

closure of $N=4$ superconf. algebra $\rightarrow E$ is a conf. killing spinor $E(x) = E_0 + x^\mu \Gamma_\mu E_5$ [Peshun 07, app. B]

[Dymarsky Peshun 09]

$N=4$ superspace eg [Drummond, Henn, K, S, 08, app B]
ladders only

$$\langle W_{\square}[\text{circle}] \rangle = \frac{2}{\sqrt{\lambda}} I_1(\sqrt{\lambda}), \quad N \gg 1$$

Peshun's localisation

$$\langle W_R[\text{circle}] \rangle_{\mathbb{R}^4} = \langle \cdot \rangle_{\mathbb{S}^4} = \left\langle \frac{1}{d(R)} \text{tr}_R e^{2\pi M} \right\rangle_{m,m}$$

3.3 Exact rev from m.m

$$Z = \int dM \exp\left(-\frac{8\pi^2}{\lambda} \text{tr} M^2\right)$$

1. Diagonalize M $Z = \int \prod_i dm_i e^{N^2 S_{\text{eff}}(m_i)}$

2. $N \gg 1$, saddle pt. expansion

$$0 = \frac{8\pi^2}{\lambda N} m_i - \frac{1}{N} \sum_{j \neq i} \frac{1}{m_i - m_j}$$

$$0 = \frac{8\pi^2}{\lambda} m = \oint \frac{\rho(m') dm'}{m - m'}$$

3. Define resolvent

$$w(m) = \oint \frac{\rho(m') dm'}{m - m'}$$

- analytic in m except in I

- $m \rightarrow \infty$, $w \sim \frac{1}{m}$

- discontinuity eg

$$w(m+i\epsilon) - w(m-i\epsilon) = -2\pi i \rho(m)$$

Find w

$$\rightarrow \sum_i \frac{1}{m_i - m} \quad (\text{eqn})$$

$$\frac{8\pi^2}{\lambda} (1 - mw(m)) = -\frac{1}{2} w(m)^2 \Rightarrow w(m) = \frac{8\pi^2}{\lambda} \left(m - \sqrt{m^2 - \frac{\lambda}{4\pi^2}} \right)$$

4. Discontinuity eq comes from $\sqrt{}$

$$\rho(m) = \begin{cases} 0 & m \notin I \\ -\frac{1}{2\pi i} \frac{8\pi^2}{\lambda} \left(-2 \sqrt{m^2 - \frac{\lambda}{4\pi^2}} \right) = \frac{8\pi^2}{\lambda} \sqrt{\frac{\lambda}{4\pi^2} - m^2} & m \in I \end{cases}$$

Renormalization, μ [Korchemsky, Radyushkin 87]

RG eq. for single cusp WL $\overbrace{\text{anomalous dim. in Callan-Symanzik equation}}$

$$\left[\mu \frac{\partial}{\partial \mu} + \beta(g_{\text{ren}}) \frac{\partial}{\partial g_{\text{ren}}} + \Gamma_{\text{cusp}}(g_{\text{ren}}, \mu, \phi) \right] W_{\text{ren}}[\text{cusp}] = 0$$

$\downarrow$ $\mu \frac{\partial}{\partial \mu} g_{\text{ren}}(\mu)$ $\downarrow$ $\mu \frac{\partial}{\partial \mu} Z_{\text{cusp}}(g_{\text{ren}}, \mu, \phi)$

for n-pt function of local ops

eq. $\rightarrow$ exponentiation of Γ_{cusp}

Γ_{cusp} in SYM [3 papers by Correa, Maldacena, Sever 12]

$\Gamma_{\text{cusp}}(\phi, \theta, \lambda)$ [Zarembo 16]

- IR divergences of scattering amplitudes of massive coloured particles

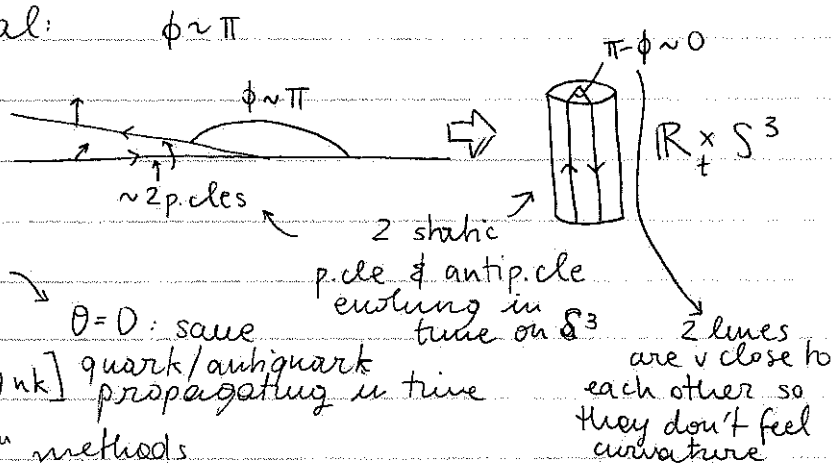
- $\theta = 0, \phi = i\varphi \rightarrow i\infty \quad \Gamma_{\text{cusp}}(\lambda, i\varphi, 0) \sim \frac{\varphi}{2} f(\lambda)$

[Beisert, Eden, Staudacher 07]

- quark-antiquark potential:

$$\Gamma_{\text{cusp}} = \frac{V(\lambda, \theta)}{\pi - \phi}$$

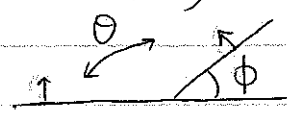
$$V(\lambda, \phi, \theta) = \frac{V(\lambda, \theta)}{L}$$



[Gromov, Levkovich-Maslyuk] quark/antiquark propagating in time

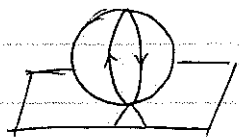
16 $\leftarrow$ using "analytic" methods

WL non locally BPS - cannot apply localisation

- ~~WL~~ $\theta = \pm \phi$ 

BPS configuration $\Gamma_{\text{cusp}}(\lambda, \phi, \pm \phi) = 0$

DGRT on S^2



near-BPS limit: $\Gamma_{\text{cusp}}(\lambda, \phi, \theta) \sim (\theta^2 - \phi^2) B(\lambda)$

$\phi \sim \theta \sim 0$

Bremsstrahlung fn $\rightarrow$

[Drucker 72], [Correa, Mald., Sever 12] [H + Henn 12]

TBA eqs. loc.